

Two Proof Technologies for the Two-Families Theorem: Permutations, Exterior Powers, and General Position in Lean 4

Louis Gestin
Independent student, France
louis@gestin.net

Independent technical report, July 2026

Abstract

We report a Lean 4 / Mathlib formalization of the Bollobás–Lovász–Frankl circle of two-families theorems, developed along two mathematically distinct proof technologies. The first is a permutation-counting proof of Bollobás’s weighted set-pairs inequality [2] and its uniform corollary. The second is an exterior-algebra proof of the Lovász–Frankl skew two-families theorem for subspaces [6], bridged to the Frankl–Kalai skew theorem for finite sets [4, 5] by a moment-curve general-position construction over $\mathbb{Q}$. All five principal theorems are fully machine-checked, each depending only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`; the development contains no `sorry`, `admit`, or custom axiom. The code comprises roughly 1440 lines across 13 files (51 declarations, 37 of them theorems), builds cleanly in about 133s when the pinned Mathlib cache is already available. Our thesis is that although the two proofs use disjoint libraries, both produce the same abstract object—a triangular certificate of independence—and we isolate several reusable components toward Mathlib: a triangular linear-independence principle, vanishing/nonvanishing and concatenation lemmas for decomposable exterior vectors, an exact division-free separation-event count, and a moment-curve general-position theorem with span-intersection consequences. The work is a formalization and introduces no new mathematics.

1 Introduction

Two-families theorems are cornerstones of extremal combinatorics. In their simplest uniform form they assert: if $A_1, \dots, A_m$ are a -sets and $B_1, \dots, B_m$ are b -sets with $A_i \cap B_i = \emptyset$ and $A_i \cap B_j \neq \emptyset$ for $i \neq j$, then $m \leq \binom{a+b}{a}$. Bollobás’s weighted refinement [2] drops the uniformity and gives the exact rational bound $\sum_i 1 / \binom{|A_i| + |B_i|}{|A_i|} \leq 1$. Lovász [6] replaced sets by subspaces and proved a *skew* version (cross-intersection required only for $i < j$), whose natural proof uses exterior algebra; Frankl [4] and Kalai [5] transported the skew bound back to finite sets via a general-position representation. Textbook expositions appear in Babai–Frankl [1].

These proofs exhibit two genuinely different technologies. The weighted inequality is proved by counting, over all linear orders of the ground set, the “separation events” in which each A_i entirely precedes B_i ; disjointness of these events across indices yields the bound. The subspace theorem is proved by attaching to each pair a decomposable exterior vector and a multiplication map, and invoking a triangular linear-independence argument inside $\bigwedge^a V$, whose dimension is exactly $\binom{a+b}{a}$. We formalize both, in Lean 4 [3] on top of Mathlib [7].

Thesis. Our central observation is that the two proofs, despite using disjoint parts of the library, converge on the same abstract certificate: a family of objects that are “nonzero on the diagonal and vanishing strictly below it.” In the counting proof this is a family of pairwise disjoint separation events of exactly computed size; in the algebraic proof it is a family of exterior-multiplication test maps feeding a single generic lemma `LinearIndependent.of_upperTriangular_maps`. Formalization makes the correspondence precise rather than metaphorical.

Contributions.

1. A complete, kernel-checked Lean development of the weighted and uniform Bollobás theorems, the Lovász–Frankl subspace theorem, the moment-curve general-position theorem, and the Frankl–Kalai skew set theorem, along two independent proof routes (Sections 4–6).
2. Reusable infrastructure for which we did not locate a direct equivalent in Mathlib v4.28.0: triangular independence from test maps; exterior decomposable nonvanishing and concatenation; an exact, division-free count of order-separating enumerations; and moment-curve general position with span-intersection lemmas (Section 7).
3. An independent semantic audit, adversarial hypothesis testing, and a measured evaluation (Section 8), supporting cautious novelty claims (Section 9).

2 Mathematical overview

Throughout, α is a finite ground type and ι a finite index type; $n = |\alpha|$.

Theorem 1 (Weighted Bollobás). *For $A_i, B_i \subseteq \alpha$ with $A_i \cap B_i = \emptyset$ for all i and $A_i \cap B_j \neq \emptyset$ for all $i \neq j$,*

$$\sum_i \frac{1}{\binom{|A_i|+|B_i|}{|A_i|}} \leq 1.$$

Corollary 2 (Uniform Bollobás). *If additionally $|A_i| = a$ and $|B_i| = b$ for all i , then $|\iota| \leq \binom{a+b}{a}$, and the bound is attained by the complement family on $\{1, \dots, a+b\}$.*

Theorem 3 (Lovász–Frankl, subspaces). *Let $\dim V = a + b$ and $U_i, W_i \leq V$ with $\dim U_i = a$, $\dim W_i = b$, $U_i \cap W_i = 0$, and $U_i \cap W_j \neq 0$ for $i < j$. Then $m \leq \binom{a+b}{a}$.*

Theorem 4 (Frankl–Kalai, sets). *Let $A_i, B_i \subseteq \alpha$ with $|A_i| = a$, $|B_i| = b$, $A_i \cap B_i = \emptyset$, and $A_i \cap B_j \neq \emptyset$ for $i < j$. Then $m \leq \binom{a+b}{a}$.*

Short informal proofs. For Theorem 1, a uniformly random linear order of α separates (A_i, B_i) with “probability” $1/\binom{|A_i|+|B_i|}{|A_i|}$; distinct separation events are disjoint (a crossing pair forces contradictory orderings), so the reciprocals sum to at most 1. We formalize this by an *exact integer count* rather than a probability. For Theorem 3, decomposable vectors $u_i = \bigwedge U_i \in \bigwedge^a V$ and multiplication by $w_j = \bigwedge W_j$ give maps with $u_i \wedge w_i \neq 0$ (transversal diagonal) and $u_i \wedge w_j = 0$ for $i < j$ (a nonzero intersection makes the combined family dependent); triangularity yields independence of the u_i , and $m \leq \dim \bigwedge^a V = \binom{a+b}{a}$. Theorem 4 follows by realizing sets as moment-curve spans, which turns $\cap$ -emptiness into $\oplus$ and $\cap$ -nonemptiness into a shared nonzero vector.

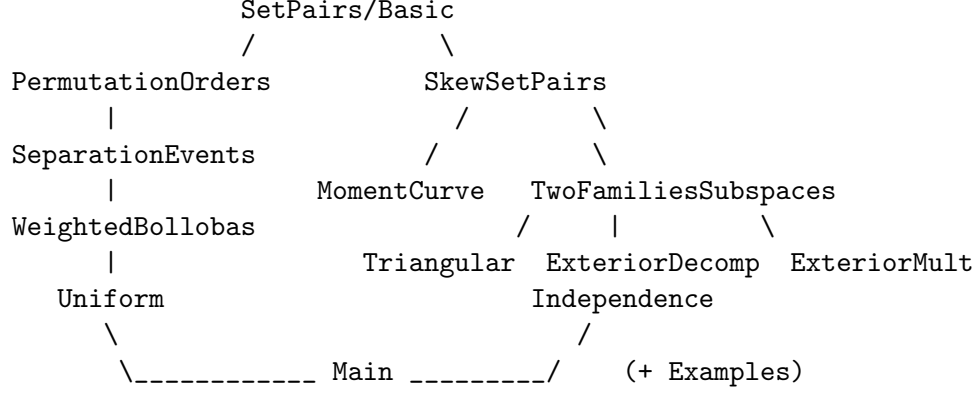


Figure 1: Project module dependency graph. Left branch: permutation counting. Right branch: exterior powers and general position. The two branches meet only at `SetPairs/Basic` (predicates) and at `Main`.

3 Formalization architecture

The development is modular and splits into two routes that share only the predicate API `SetPairs.Basic`. Figure 1 shows the project-local import DAG (generated from the `import RequestProject.*` lines; verified acyclic in `DEPENDENCY_AUDIT.md`).

- `SetPairs/:` `Basic` (predicates), `PermutationOrders`, `SeparationEvents`, `WeightedBollobas`, `Uniform`.
- `LinearAlgebra/:` `TriangularIndependence`, `ExteriorDecomposable`, `ExteriorMultiplication`, `TwoFamiliesSubspaces`, `MomentCurve`.
- `SkewSetPairs`, `Examples`, `Main`.

4 Exact permutation counting

An *order enumeration* of α is an equivalence $e : \alpha \simeq \text{Fin } n$; there are $n!$ of them (`card_orderEnumerations`, from `Fintype.card_equiv`). The pair (A, B) is *separated* by e when every element of A precedes every element of B :

```
def Separates (e : OrderEnumeration a) (A B : Finset a) : Prop :=
  forall {x}, x in A -> forall {y}, y in B -> e x < e y
```

and `separationEvent(A, B)` is the finset of separating enumerations.

The exact count. The heart of the argument is

$$|\text{separationEvent}(A, B)| \cdot \binom{|A| + |B|}{|A|} = n!,$$

which we obtain from the closed form (`card_separationEvent_eq`)

$$|\text{separationEvent}(A, B)| = \binom{n}{a+b} a! b! (n - (a+b))!,$$

with $a = |A|$, $b = |B|$, $S = A \cup B$. This count is proved by a genuine *ordering-factorization* equivalence: a separating enumeration is determined by three independent pieces:

- the $(a+b)$ -element position set $P = e(S) \subseteq \text{Fin } n$ (contributing $\binom{n}{a+b}$);
- the relative order of S inside P , in which A precedes B (contributing $a!b!$);
- the relative order of S^c inside P^c (contributing $(n - a - b)!$).

Concretely:

- `card_separates_full`: when $A \cup B = \text{univ}$, the separating enumerations biject with $(A \simeq \text{Fin } a) \times (B \simeq \text{Fin } b)$, giving $a!b!$. The bijection is built from two explicit injections whose well-definedness rests on the order-statistics facts that separation forces every A -element below rank a (`separates_val_lt`) and every B -element at rank $\geq a$ (`card_le_separates_val`).
- `card_separates_toP`: transports this count along the order isomorphism $P \simeq_o \text{Fin}(a+b)$ to an arbitrary ordered position set P .
- `fiber_card_eq`: for a fixed P , the enumerations with $e(S) = P$ biject with $(\text{relative orders of } S) \times (S^c \simeq P^c)$.
- `card_separationEvent_eq`: sums the fibre counts $a!b!(n - (a+b))!$ over the $\binom{n}{a+b}$ position sets, via `Finset.card_eq_sum_card_fiberwise`.

The factorial identity then follows *without any natural-number division*, from $\binom{a+b}{a}a!b! = (a+b)!$ and $\binom{n}{a+b}(a+b)!(n - (a+b))! = n!$ (`Nat.choose_mul_factorial_mul_factorial`, applied twice).

Why not probability. A direct probabilistic phrasing would introduce $\mathbb{R}$ -valued measures and division, whose formal handling is heavier than the integer identity above and obscures the fibre structure that is the real proof. The integer count is also what the regression tests can check exactly (Section 8).

Disjointness and assembly. Separation events are pairwise disjoint: if $x \in A_i \cap B_j$ and $y \in A_j \cap B_i$, then separating (A_i, B_i) forces $e(x) < e(y)$ while separating (A_j, B_j) forces $e(y) < e(x)$ (`separationEvents_disjoint`). This step is exactly where cross-intersection in *both* directions is used. Summing the disjoint cardinalities gives $\sum_i |E_i| \leq n!$, and since $1/\binom{|A_i|+|B_i|}{|A_i|} = |E_i|/n!$ the weighted inequality (`weighted_bollobas`) follows. The uniform corollary (`uniform_bollobas`) factors out the constant reciprocal and divides; sharpness is the complement family on $\text{Fin}(a+b)$ (`uniform_bollobas_sharp`).

5 Triangular independence and exterior products

Triangular independence. If linear maps T_j satisfy $T_j(u_j) \neq 0$ and $T_j(u_i) = 0$ for $i < j$, then the u_i are linearly independent (`LinearIndependent.of_upperTriangular_maps`):

```
theorem LinearIndependent.of_upperTriangular_maps
  (u : Fin m -> M) (T : Fin m -> M ->L[K] N)
  (hdiag : forall j, T j (u j) != 0)
  (hzero : forall {i j}, i < j -> T j (u i) = 0) :
  LinearIndependent K u
```

The proof takes a vanishing combination $\sum_i g_i u_i = 0$, lets j be the *largest* index with $g_j \neq 0$, and applies T_j : terms with $i < j$ die by `hzero`, terms with $i > j$ die by maximality, leaving $g_j T_j(u_j) = 0$, a contradiction since K is a field. The maximal-index choice is matched to the upper-triangular orientation; a dual of `lowerTriangular_maps` uses the minimal index.

Decomposable vectors. The alternating product $\iota \text{Multi } K r v$ vanishes exactly when v is linearly dependent: the vanishing direction is `AlternatingMap.map_linearDependent`, and the nonvanishing direction (`Multi.ne_zero_of_linearIndependent`, both in the exterior power and the exterior algebra) is obtained from Mathlib’s `Multi.family_linearIndependent_field` applied to the unique full subset. Concatenation $\iota \text{Multi } a u \cdot \iota \text{Multi } b w = \iota \text{Multi } (a+b) (u ++ w)$ (`Multi.mul_Multi`) turns products of decomposables into the decomposable of the appended family; its proof is an element-wise `List.ofFn/Fin.append` identity.

Subspace theorem. For each i we pick bases of U_i and W_i (via `Module.finBasisOfFinrankEq`, using $\dim U_i = a$, $\dim W_i = b$), forming $u_i \in \bigwedge^a V$ and a test map $T_j = (\cdot * w_j) \circ \iota$ where w_j is the decomposable of W_j . Then $T_j(u_i)$ is the decomposable of the concatenated bases, nonzero on the diagonal (since $U_i \cap W_i = 0$ makes the union independent, `append_wedge_ne_zero`) and zero for $i < j$ (since $U_i \cap W_j \neq 0$ makes it dependent, `append_wedge_eq_zero`). Triangular independence bounds $m \leq \dim \bigwedge^a V = \binom{a+b}{a}$ via `exteriorPower.finrank_eq`. The degenerate cases $a = 0$, $b = 0$, $m = 0$ require no special handling: empty bases and $\iota \text{Multi } K 0 = 1$ are absorbed uniformly—a case split that appeared necessary on paper turned out to be an artifact of informal reasoning.

6 Moment-curve bridge

The moment-curve vector is $v_x = (1, t_x, \dots, t_x^{d-1}) \in \mathbb{Q}^d$ for an injective label $t : \alpha \hookrightarrow \mathbb{Q}$. Any at most d distinct such vectors are linearly independent (`momentCurve_linearIndependent`); we prove this by polynomial interpolation—the polynomial $\prod_{j \neq i} (X - t_j)$, of degree $|S| - 1 < d$, isolates index i —rather than by expanding the Vandermonde determinant, which is cleaner in Lean and, crucially, handles the *rectangular* case $|S| < d$ directly without any square-matrix reduction. As consequences we obtain the span dimension $\dim \text{span}\{v_x : x \in S\} = |S|$ for $|S| \leq d$ (`finrank_spanMomentCurve`), triviality of $\text{span}(S) \cap \text{span}(T)$ for disjoint S, T with $|S| + |T| \leq d$ (`inf_spanMomentCurve_eq_bot`), and nontriviality when $S \cap T \neq \emptyset$ (`inf_spanMomentCurve_ne_bot`, using that each v_x is nonzero because its zeroth coordinate is 1). The Frankl–Kalai theorem then labels the ground type injectively into $\mathbb{Q}$ (via `Fintype.equivFin` composed with $\mathbb{N} \hookrightarrow \mathbb{Q}$), realizes each set as a moment-curve span in $\mathbb{Q}^{a+b}$, and applies the subspace theorem. The ground set may be strictly larger than $a + b$; only the ambient dimension $d = a + b$ is constrained.

7 Reusable infrastructure and Mathlib gaps

Table 1 maps the central mathematical objects to their Lean representations. Several components are generic and were not located in Mathlib v4.28.0 during our documented search (`MATHLIB_PR_PLAN.md`): triangular independence from test maps; exterior decomposable nonvanishing and concatenation (Mathlib has only the vanishing direction); the exact separation-event count; and moment-curve general position with span-intersection lemmas. Mathlib does supply the exterior-power API and its dimension $\dim \bigwedge^a V = \binom{\dim V}{a}$ (`exteriorPower.finrank_eq`), the Vandermonde determinant, and the LYM inequality (a different chain-counting technique).

Mathematical object	Lean representation	Key declaration	Main difficulty
Total ordering	$\alpha \simeq \text{Fin } n$	<code>OrderEnumeration</code>	counting bijections
Separation event	<code>filtered Finset</code>	<code>separationEvent</code>	exact fibre count
Exterior decomposable	$\iota \text{Multi } K \text{ } r \text{ } v$	$\iota \text{Multi_ne_zero_}$ *	nonvanishing direction
Exterior mult. test	<code>mulRight</code> $\circ$ subtype	$\iota \text{Multi_mul_}$ ιMulti	append identity
Moment-curve vector	$\text{Fin } d \rightarrow \mathbb{Q}$	<code>momentCurve</code>	rectangular independence
Set span	<code>Submodule.span</code>	<code>spanMomentCurve</code>	$\cap$ vs. $\oplus$
Skew cross-intersection	$i < j \rightarrow (A_i \cap B_j) \neq \emptyset$	<code>SkewCrossIntersecting</code>	orientation

Table 1: Informal-to-Lean correspondence (Table A).

Theorem	Method	Lean declaration	Checked	Sharp	Reusable deps
Weighted	counting	<code>weighted.bollobas</code>	yes	—	sep. count
Uniform	counting	<code>uniform.bollobas</code>	yes	yes	weighted
Triangular	lin. alg.	<code>of_upperTriangular_maps</code>	yes	—	(generic)
Subspace	exterior	<code>lovasz_frankl_subspaces</code>	yes	—	triangular, exterior
Moment curve	interp.	<code>momentCurve_linearIndependent</code>	yes	—	(generic)
Frankl–Kalai	bridge	<code>frankl_kalai_skew</code>	yes	yes	subspace, moment

Table 2: Theorem coverage (Table B). “Sharp” = a matching extremal family is formalized; the uniform/skew bound is attained by the complement construction.

8 Evaluation

Versions. Lean v4.28.0; Mathlib pinned at v4.28.0 (commit 8f9d9cff). Project clean build with the pinned Mathlib artifacts already cached: **success, 8039 jobs, ≈ 133 s** wall-clock in our environment. A cold reconstruction first downloads the pinned dependencies and obtains the Mathlib cache, so it takes substantially longer.

Size. 13 Lean files, 1438 total lines (1175 non-blank), 51 declarations: 37 theorems/lemmas, 10 definitions, 2 abbreviations, 2 instances. By module the largest is `SeparationEvents.lean` (481 lines), followed by `MomentCurve.lean` (179) and `TwoFamiliesSubspaces.lean` (144).

Axioms. Every principal theorem’s `#print axioms` lists exactly `[propext, Classical.choice, Quot.sound]`; no `sorryAx`. The repository contains no `sorry`, `admit`, `custom axiom`, `unsafe`, or `@[implemented_by]` (verified by `scripts/check_forbidden.sh`).

Regression tests. `scripts/verify_small_cases.py` exhaustively checks, in exact rational arithmetic: the weighted inequality on ground sets of size 2–5; the uniform bound for admissible (a, b) on ground sets of size ≤ 6 ; the skew bound on ground sets of size ≤ 4 ; and the sharpness construction (OVERALL: PASS). `scripts/adversarial_checks.py` confirms with exact counterexamples that each principal hypothesis is load-bearing (ADVERSARIAL_TESTS.md). Small Lean `decide/norm_num` checks (`Examples.lean`) corroborate specific binomial values, the closed-form count $\binom{n}{a+b} a! b! (n - (a+b))!$ over five boundary cases (empty A/B , elements outside $A \cup B$, and $A \cup B = \text{univ}$), and two adversarial sums exceeding 1. The computational checks are regression tests, not substitutes for the proofs.

Reusable fraction. The generic infrastructure (Sections 5–7: triangular independence, exterior nonvanishing/concateration, moment-curve general position, separation-event counting) is application-independent and constitutes the bulk of the extraction candidates in `MATHLIB_PR_PLAN.md`.

Failed and revised approaches. Moment-curve independence was first attempted through the rectangular Vandermonde matrix, then replaced by the cleaner polynomial-interpolation argument. The subspace theorem was initially planned with explicit case splits on $a = 0$, $b = 0$, $m = 0$; these proved unnecessary once the argument was routed through the exterior algebra directly. The separation-event count was initially phrased with natural-number division and re-derived as an exact product identity to avoid divisibility side-conditions.

9 Related work

The mathematical originals are Bollobás [2] (weighted/uniform set-pairs), Lovász [6] (exterior-algebra subspace method), and Frankl [4]/Kalai [5] (skew set extensions), with the exposition of Babai–Frankl [1]. Mathlib [7] provides the exterior-power and graded-exterior-algebra API, the Vandermonde determinant, and the LYM inequality, but—to the best of our documented search (`NOVELTY_AUDIT.md`)—no Bollobás set-pairs inequality, uniform/skew two-families theorem, or Lovász subspace theorem. The local Mathlib source was searched directly, while external indexes and libraries were surveyed to the extent available in the audit environment; a live re-check remains advisable before making a stronger novelty claim. We did not locate a prior formalization combining the weighted permutation proof, the Lovász–Frankl subspace theorem, and the moment-curve reduction to the Frankl–Kalai skew theorem in one development, and make no stronger claim (in particular not “the first formalization” of any individual theorem, since live external search was unavailable).

10 Lessons and limitations

Lessons. (i) Exact integer identities beat probabilistic phrasing in a proof assistant, both for the proof skeleton and for exact regression testing. (ii) Setting up the algebra uniformly can make degenerate cases dissolve rather than require case splits. (iii) Rectangular general position must be genuinely proved (via interpolation), not assumed by analogy with the square Vandermonde matrix. (iv) Subtype/cardinality transport (between $\uparrow S$, $\text{Fin } |S|$, and position sets) is a recurring cost centre in the counting proof, handled by explicit order isomorphisms.

Limitations. The development relies on classical choice and finite ground and index types. Equality/extremal classification beyond the single complement construction is not formalized. The moment-curve layer is stated over $\mathbb{Q}$ rather than an arbitrary (infinite) field. As the entire project workflow was substantially AI-assisted—including planning, proof development, code, testing, writing, translation, auditing, and repository preparation—kernel-checking guarantees correctness only *relative to the stated theorems*; semantic fidelity is argued separately (`INDEPENDENT_PROOF_AUDIT.md`, `SEMANTIC_AUDIT.md`) and every claim remains the responsibility of the human author (`AI_USE_STATEMENT.md`).

11 Conclusion

We have formalized, along two distinct proof technologies, the Bollobás–Lovász–Frankl two-families circle in Lean 4—the weighted and uniform Bollobás theorems by permutation counting, and the Lovász–Frankl subspace and Frankl–Kalai skew set theorems by exterior powers and moment-curve general position—together with reusable algebraic and combinatorial infrastructure. All five principal theorems are kernel-checked under the standard axioms. Beyond the specific theorems, the development isolates a common triangular-certificate structure underlying two superficially unrelated proofs, and packages several components in Mathlib-ready form. The work claims no new mathematics.

Artifact availability

The complete Lean sources, scripts, audit documents, LaTeX sources, and both language versions of this report are distributed in the accompanying project archive. The development is pinned to Lean v4.28.0 and Mathlib v4.28.0; the principal build command is `lake build`.

References

- [1] László Babai and Peter Frankl. Linear algebra methods in combinatorics, 1992. Preliminary version, Department of Computer Science, University of Chicago.
- [2] Béla Bollobás. On generalized graphs. *Acta Mathematica Academiae Scientiarum Hungaricae*, 16(3–4):447–452, 1965.
- [3] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Proceedings of the 28th International Conference on Automated Deduction (CADE)*, pages 625–635, 2021.
- [4] Peter Frankl. An extremal problem for two families of sets. *European Journal of Combinatorics*, 3(2):125–127, 1982.
- [5] Gil Kalai. Intersection patterns of convex sets. *Israel Journal of Mathematics*, 48(2–3):161–174, 1984.
- [6] László Lovász. Flats in matroids and geometric graphs. In *Combinatorial Surveys (Proc. Sixth British Combinatorial Conf., Royal Holloway Coll., Egham, 1977)*, pages 45–86. Academic Press, 1977.
- [7] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP)*, pages 367–381, 2020.